

Cecil Cao . Jarvis Consulting

I am a Computer Science graduate from the University of Toronto with 3+ years of experience in application and systems development in various tech stacks, such as React/Typescript, Java, and Python. At Jarvis, I have been responsible for developing backend systems and databases, running data analytics, and business solutions; the goal being to reduce work hours through automation. I have also been part of a research team exploring ways to integrate robots into the laboratory and for scientists to manage multiple labs through virtual reality remotely. I enjoy the process of optimizing work, either through automation or using project management tools, and always seek to make repetitive work a little easier.

Skills

Proficient: C#, Java, Python, Linux/Bash, RDBMS/SQL, Unity

Competent: C++, Git, Cloud Computing (AWS, GCP), Docker, PostgreSQL, Agile Methodologies

Familiar: Project Management Tools (Notion, Jira), GraphQL, Golang, Blender, IsaacSim/IsaacLab

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_CecilCao

Cluster Monitor [GitHub]: Deployed an automated grep app using Docker and stored the data in a PostgreSQL database. Generated Bash scripts to grep physical load across a closed cluster running Rocky Linux OS and automated with a CronJob.

Core Java Apps [GitHub]: Programmed a grep log reader to handle large log files in Java using basic OOP concepts, such as polymorphism using interfaces and overriding. Utilized the JUnit library for unit testing and slf4j for logging and debugging.

Python Data Analytics [GitHub]: Analyzed a sanitized dataset from the London Gift Shop with PostgreSQL and Jupyter, and presented insights on their types of customers, as well as their geographic outreach to increase revenue growth.

BSA Fundamentals [GitHub]: Drafted a software requirement document outlining the hardware, software, and development constraints and 8-month scope to develop a robust market order trading app for the Bank of Jarvis' online service products.

BSA Elicitation Project [GitHub]: Presented a business solution to a managing director and the team for an automated dashboard in order to minimize human error during the document drafting process. Drafted a business requirements document to outline the requirements, constraints, and timeline by an elicitation with the managing director, as well as, the rest of the team for clarification on requirements and to manage the needs of everyone on the team.

Highlighted Projects

Into the Dataswarm (Beta) [GitHub]: Managed the Trello kanban board and Git repository, ran weekly Agile sprints to break down key milestones into actionable tickets, prioritized bug fixes and feature tickets each sprint. Consulted with playtesters and the game's creative director in order to realize and refine stakeholder goals.

Toronto Job Market Database: Discovered COVID-19's impact on the Toronto job market using the open datasets and determined job sectors affected most by the pandemic. Utilized Python (Pandas and Pyplot) to set up an ETL pipeline onto a PSQL database.

Professional Experiences

Business Systems Analyst, Jarvis (2026-present): Working with key executives and stakeholders to manage sprints by keeping tasks organized in Notion. Writing validation scripts in Python to support the team's workflow. Providing updates and resolving daily blockers to maintain steady progress.

Research Assistant, University of Toronto (2025-2026): Collaborated with a VR simulation for a robotics lab to automate chemical experiments in a wet lab with robots performing the experiment using IsaacLab and Python to allow scientists to walk through a virtual lab and control a robot.

Junior Software Engineer, Alida Inc. (2022-2023): Collaborated with a professional team to improve upon a full-stack website with React, Typescript, Golang and PSQL. Improved upon their automated form validation for better error checking.

Education

University of Toronto (2018-2025), Bachelor of Computer Science, Arts and Science

Miscellaneous

- Hosts creative writing sessions among peers and writes flash fiction pieces for Role-playing Games.
- An active member of the International Game Developers Association, and developing an indie game.
- Tutors language arts and mathematics to K12 students on the weekends.